BSPD CALCULATIONS
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All voltages are measured from GND.
+5V comes from the L7805. Nominal is 5.00 V, but it can be anywhere from 4.80 V to 5.20 V.

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1. POWER

The L7805 (U4) takes +12V GLV from J1 and makes +5V.
Input caps: C1 10uF, C2 100nF
Output caps: C3 10uF, C4 100nF
Each MCP6004 has its own 100nF cap (C6, C10, C11).

Current on +5V:

HASS 50-S	25 mA (max)
AiM pressure sensor	15 mA (estimate, need to check datasheet)
3x MCP6004	12 amps x 0.17 mA = 2.0 mA
divider resistors	about 1.0 mA
R26 pull-up	0.5 mA
total	about 44 mA

7805 power = (14.4 V - 5 V) x 0.044 A = 0.41 W
That is low enough that it doesn't need a heatsink.

Current on +12V = 44 mA + relay coil 17 mA + 7805 itself 5 mA = about 66 mA

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2. BRAKE PRESSURE

Sensor: AiM X05PSA02000P18, 0-2000 psi, 0.5-4.5 V output
V = 0.5 + 0.002 x psi

0 psi -> 0.50 V
435 psi -> 1.37 V (30 bar, placeholder)
1000 psi -> 2.50 V
2000 psi -> 4.50 V

R1 = 100k pull-down, so if the sensor is unplugged the signal goes to 0 V.

Filter R2 / C5:
  time constant = 10k x 100nF = 1 ms
  cutoff = 1 / (2 x pi x 10k x 100nF) = 159 Hz

Brake threshold (BRAKE_REF):
  R3 is 10k from +5V to the top of RV1 (10k pot), so the top of the pot is
  5 x 10k / (10k + 10k) = 2.50 V
  The wiper can be set from 0 to 2.50 V, which is 0 to 1000 psi.
  For 30 bar: set BRAKE_REF = 1.37 V

The sensor and the reference are both powered from the same +5V, so if +5V changes the pressure where it trips stays the same.

U1A: + is BRAKE_FILT, - is BRAKE_REF, output is BRAKE_HIGH
  BRAKE_HIGH = 5 V when the brake is above the threshold

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3. CURRENT

Sensor: LEM HASS 50-S, 50 A nominal, +/-150 A range
  Uout = Uref + 0.0125 x amps (12.5 mV per amp)
  Uref = 2.50 V, made inside the sensor (does NOT follow +5V)

  -150 A -> 0.625 V
    0 A -> 2.500 V
  +150 A -> 4.375 V

Load on the HASS output:
  R4 (10k) in parallel with R5 + R6 (59.9k) = 8.57k
  Datasheet minimum is 2k, so this is fine.
  C8 is behind R10 (10k), so the sensor doesn't see the 100nF directly.

Uref buffer:
  U3B is a voltage follower. HASS_REF goes in, REF_BUF comes out.
  R27 = 100k pull-down on HASS_REF.
  Normal: 2.5 V / 100k = 25 uA, and 25 uA x 200 ohm = 5 mV error (tiny)
  If the Uref wire breaks, HASS_REF goes to 0 V and U3C catches it.

Difference amp (U1B):
  R5 = R7 = 10k, R6 = R8 = 49.9k

gain = 49.9k / 10k = 4.99
I_SCALED = 4.99 x (HASS_OUT - REF_BUF)
= 4.99 x 0.0125 x amps
= 0.0624 V per amp

0 A gives 0 V. The output maxes out around 4.97 V, which is about 80 A.

Filters:
R9 / C7 on I_SCALED: 10k x 100nF = 1 ms
R10 / C8 on HASS_OUT: 10k x 100nF = 1 ms

Current threshold:
rule limit = 5000 W / 108 V = 46.3 A

The original R12 (13.7k) put the threshold right at 46.3 A. But the
divider comes from +5V and the HASS doesn't, so if the 7805 is at 5.2 V
the trip point moves up to 48.2 A = 5.2 kW, which is over the limit.
I changed R12 to 11.0k to leave room.

R11 = 10k, R12 = 11.0k 1%
CURRENT_REF = 5 x 11.0 / (10 + 11.0) = 2.619 V
trip current = 2.619 / 0.0624 = 42.0 A
trip power = 42.0 x 108 = 4.53 kW

at 4.80 V supply: 2.514 V -> 40.3 A -> 4.35 kW
at 5.20 V supply: 2.724 V -> 43.7 A -> 4.72 kW

Even with 1% resistor error in the difference amp, the worst case is
about 44.6 A = 4.8 kW, which is still under 5 kW.

U1C: + is I_FILT, - is CURRENT_REF, output is POWER_HIGH

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4. AND (U1D)

R13 and R14 (both 10k) average BRAKE_HIGH and POWER_HIGH into SUM:
SUM = (BRAKE_HIGH + POWER_HIGH) / 2

neither on -> 0 V
one on -> 2.5 V
both on -> 5 V

AND_REF = 5 x 30k / (10k + 30k) = 3.75 V

Only "both on" (5 V) is above 3.75 V. There is 1.25 V of margin either way.

U1D: + is SUM, - is AND_REF, output is BOTH_HIGH

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5. TIMER (R17, C9, U2A)

BOTH_HIGH charges C9 through R17. U2A trips when C9 reaches TIMER_REF.

TIMER_REF = 5 x 10k / (10k + 10k) = 2.50 V (half of +5V)

Charging to half of the voltage always takes R x C x ln(2), so the
time doesn't depend on the supply:

t = R17 x C9 x 0.693

old: 680k x 1uF x 0.693 = 0.471 s
with a +/-10% cap: 0.42 to 0.52 s -> could go over 0.5 s
new: 470k x 1uF x 0.693 = 0.326 s
with a +/-10% cap: 0.29 to 0.36 s

The rule says it has to trip if both happen for more than 0.5 s, so 0.5 s is
the longest allowed. 0.33 s leaves room for tolerance.

C9 should be 1uF X7R, 16 V or higher.

If the condition stops early, C9 drains back through R17 (time constant
0.47 s). Short events close together add up, so it can trip early. That
is the safe direction.

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6. LATCH AND RESET (D1, R20)

When TRIP goes high it feeds back into C9 through D1 and R20:

TRIP 5.0 V - D1 0.7 V = 4.3 V through R20 (10k)
R17 (470k) pulls the other way toward BOTH_HIGH = 0 V
TIMER = 4.3 x 470k / (470k + 10k) = 4.21 V

4.21 V is above 2.5 V, so it stays tripped even after the brake and

current go back to normal.

Reset is a power cycle. When +5V turns off, C9 drains into the +5V rail through D9 (1N4148, anode TIMER, cathode +5V). During normal use TIMER is never above 4.3 V, so D9 doesn't do anything.

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7. SENSOR FAULTS

Window limits:

FAULT_LO_REF = 5 x 10k / (182k + 10k) = 0.260 V
FAULT_HI_REF = 5 x 182k / (10k + 182k) = 4.740 V

Normal ranges, all inside the window:

brake (AiM) 0.500 to 4.500 V (0.24 V margin each side)
HASS output 0.625 to 4.375 V (0.37 V margin each side)
HASS Uref 2.475 to 2.525 V

Fault comparators (output goes high on a fault):

U2B FAULT_LO_REF vs BRAKE_FILT brake open or shorted to GND D2
U2C BRAKE_FILT vs FAULT_HI_REF brake shorted to 5V D3
U2D FAULT_LO_REF vs HASS_FILT current open or shorted to GND D4
U3A HASS_FILT vs FAULT_HI_REF current shorted to 5V D5
U3C FAULT_LO_REF vs REF_BUF Uref open or shorted to GND D7
U3D REF_BUF vs FAULT_HI_REF Uref shorted to 5V D8

All six diodes go to FAULT, and FAULT goes to TIMER through R25 (100k).

Time for a real fault to latch:

FAULT = 5.0 V - about 0.5 V diode drop = 4.5 V
R17 (470k) pulls TIMER toward 0 V at the same time, so:
effective voltage = 4.5 x 470k / (100k + 470k) = 3.71 V
effective resistance = 100k || 470k = 82.5k
time constant = 82.5k x 1uF = 82.5 ms
time to 2.5 V = 82.5 ms x ln(3.71 / (3.71 - 2.5)) = 92 ms

Power-up:

The filter caps start at 0 V, so for a moment the signals look like an open sensor.
Brake signal reaches 0.26 V in 1 ms x ln(0.5 / (0.5 - 0.26)) = 0.73 ms
C9 only charges about 45 uA x 0.73 ms / 1uF = 33 mV in that time.
This is why R25 is 100k. Without it the BSPD would latch every time the

car turns on.
Still need to check that the HASS and AiM start up in well under 90 ms.

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8. OUTPUT

Q1 (2N7002): gate = TRIP, pulls BSPD_OK low when tripped
R26 (10k): pulls BSPD_OK up to +5V
Q2 (2N7002): gate = BSPD_OK, turns the relay coil on
K1 (G6DN-1A): 12 V coil, normally open contact in the shutdown circuit (J4)
D6 (1N4148): flyback diode across the coil

normal: TRIP 0 V -> Q1 off -> BSPD_OK 5 V -> Q2 on -> relay on -> SDC closed
tripped: TRIP 5 V -> Q1 on -> BSPD_OK 0 V -> Q2 off -> relay off -> SDC open
no +5V: BSPD_OK 0 V -> Q2 off -> relay off -> SDC open
no +12V: relay off -> SDC open

2N7002 turns on fully with 5 V on the gate (threshold is 2.5 V max).
Relay coil is about 17 mA at 12 V (check G6DN datasheet). It needs about
8.4 V to turn on.
D6 keeps the voltage spike on Q2 at about 12.7 V when the relay turns off.

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9. TEST LOOP (J5, CS+ / CS-)

A separate wire goes N times through the HASS hole. It is not connected to
the circuit. Each turn adds up, so:
test current = 42.0 A / N

5 turns -> 8.4 A
10 turns -> 4.2 A
20 turns -> 2.1 A

Jacks: 4 mm shrouded banana, labeled CS+ and CS-, not red or black.
The HASS arrow has to point the way current flows out of the pack. If it is
mounted backwards, the BSPD never sees power, and the test loop would still
pass.

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10. STILL NEED TO FIGURE OUT

- what pressure counts as hard braking (30 bar is a placeholder)
- confirm 108 V nominal
- AiM connector pinout for J2
- how many test loop turns
- which switch resets the BSPD, and if a GLV power cycle counts
- peak pack current vs the HASS 150 A range, and what the HASS outputs above 150 A (if it goes over 4.74 V, U3A will trip at full power)
- HASS and AiM start-up time
- recheck against the final 2027 rules